

Unit 4: Functions Involving Parameters, Vectors, and Matrices

Name: _____ Date: _____

Period: _____

Learning Targets — I Can...

1. I can write a parametric representation of an ellipse or circle given its standard form equation by using the substitution $x - h = a \cos t$, $y - k = b \sin t$.
2. I can write a parametric representation of a parabola by introducing a parameter with the substitution $x(t) = h + t$ (or $y(t) = k + t$ for sideways parabolas).
3. I can use parametric analysis — finding extrema from component functions and axis intercepts from zeros — on a parametrized implicit curve.

What's in This Packet

#	Component	Description	Score
1	Warm-Up	Spiral review: unit circle, ellipse intercepts, evaluating parametric functions	_____
2	Guided Notes	Ellipse and parabola parametrizations; parametric analysis	_____
3	Group Activity	Mission Control: parametrize three conic paths and extract key facts	_____
4	Exit Ticket	Parametrize an ellipse; identify topmost and bottommost points	_____
5	Homework	Independent practice: ellipses, circles, parabolas, AP-style MC	_____
Total			_____